

Test of Mathematics for University Admission
Mock Set G Paper 2
 Mathematical Reasoning

Name: _____

Date sat: _____

Score: _____ / 20

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Throughout, “must be true” means true in every case allowed by the information given, and a counterexample must satisfy the hypothesis while failing the conclusion.

[illegible]

Question 1

A student solves the equation

$$\sqrt{2x+3} - \sqrt{x-2} = 2$$

by rearranging to $\sqrt{2x+3} = 2 + \sqrt{x-2}$, squaring to obtain $x+1 = 4\sqrt{x-2}$, and squaring again to reach $x^2 - 14x + 33 = 0$, so that $x = 3$ or $x = 11$. The student then writes: “Squaring can introduce solutions that do not satisfy the original equation, so the larger root must be rejected, and the only solution is $x = 3$.” Which of the following is correct?

- A The student is right: $x = 3$ is the only solution.
- B Only $x = 11$ satisfies the original equation.
- C Neither value satisfies the original equation.
- D Both $x = 3$ and $x = 11$ satisfy the original equation, so the student’s rejection is wrong.
- E The first squaring is invalid, so the quadratic is wrong.
- F There is a third solution, lost in the second squaring.

Question 2

A positive integer $n \leq 100$ is the subject of four statements:

- A: n is divisible by exactly one of the numbers 2 and 3.
- B: $n + 1$ is a perfect square.
- C: n has exactly four positive divisors.
- D: the digits of n sum to 8.

Exactly three of the four statements are true, and statement C is false. Find n .

- A 8
- B 80
- C 35
- D 62
- E 99
- F there is no such n

Question 3

For positive integers a , b and c , which of the following statements are true?

- I If a divides b^2 and a divides c^2 , then a divides bc .
 - II If a divides bc , then a divides b or a divides c .
 - III If a divides b and a divides c , then a^2 divides bc .
 - IV If a^2 divides b^2 , then a divides b .
- A I and III only
 - B III and IV only
 - C I, III and IV only
 - D I, II, III and IV
 - E III only
 - F II, III and IV only

Question 4

Real numbers a and b satisfy $a^2 + b^2 = 1$. Which of the following must be true?

- I $|a + b| \leq \sqrt{2}$
- II $a^3 + b^3 \leq 1$
- III $ab \geq -\frac{1}{2}$
- IV $a + b \geq ab$
- A I, II and III only
- B I and II only
- C I, II, III and IV
- D I and III only
- E II and IV only
- F I, III and IV only

Question 5

Which of the following statements are true for every real number x ?

- I $|x - 1| + |x + 1| \geq 2$
- II $|x - 1| - |x + 1| \leq 2$
- III $|x^2 - 1| \geq |x| - 1$
- IV $|x - 2| + |x - 5| + |x - 9| \geq 7$
- V $||x| - 3| \leq |x| - 3$
- A I and II only
- B I, II and III only
- C I, II, IV and V only
- D I, II, III, IV and V
- E I, II, III and IV only
- F III and IV only

Question 6

Three arguments are offered for the claim that $\sqrt{3}$ is irrational. In each, suppose $\sqrt{3} = \frac{p}{q}$ with p and q positive integers having no common factor, so that $p^2 = 3q^2$.

- I. $3 \mid p^2$, and 3 is prime, so $3 \mid p$. Writing $p = 3m$ gives $q^2 = 3m^2$, so $3 \mid q$ as well: a contradiction.
- II. If p were even, $3q^2$ would be even, so q would be even: a contradiction. So p is odd, and then so is q . But the square of an odd number leaves remainder 1 on division by 8, so $p^2 \equiv 1$ while $3q^2 \equiv 3 \pmod{8}$: a contradiction.
- III. $\sqrt{3} = 1.7320508\dots$ has a non-terminating decimal expansion, so it is irrational.

Which of these arguments are valid?

- A I only
- B I and II only
- C II only
- D I, II and III
- E I and III only
- F none of them

Question 7

Let n be a positive integer. Which of the following conditions are equivalent to “ n is a perfect square”?

- I n has an odd number of positive divisors.
 - II n leaves remainder 0 or 1 on division by 4.
 - III The sum of the positive divisors of n is odd.
 - IV Every prime occurs in the prime factorisation of n to an even power.
- A I only
 B IV only
 C I, II and IV only
 D I, III and IV only
 E II and IV only
 F I and IV only

Question 8

A sequence $a_1, a_2, a_3, \dots$ is called *eventually positive* if there is a number N such that $a_n > 0$ for every $n > N$. How many of the following sequences are eventually positive?

- I $a_n = n^2 - 100n$
 - II $a_n = (-1)^n + \frac{2}{n}$
 - III $a_n = \sin n + \frac{3}{2}$
 - IV $a_n = \frac{n - 1000}{n^2 + 1}$
 - V $a_n = \cos\left(\frac{n\pi}{2}\right) + 1$
- A 1
 B 2
 C 3
 D 4
 E 5

Question 9

The polynomial $x^4 + ax^3 + bx^2 + cx + d$ has four distinct positive real roots. Which of the following must be true?

- I $a < 0$ and $c < 0$
 - II $b > 0$ and $d > 0$
 - III $a^2 > 2b$
 - IV $a^2 > 3b$
- A I only
 B I and II only
 C I, II, III and IV
 D III and IV only
 E I and III only
 F I, II and III only

Question 10

A student sets out to prove that $3^n \geq n^3$ for every positive integer n .

- (I) For $n = 1$: $3 \geq 1$, which is true.
- (II) Suppose $3^k \geq k^3$ for some positive integer k .
- (III) Then $3^{k+1} = 3 \cdot 3^k \geq 3k^3$.
- (IV) And $3k^3 \geq (k+1)^3$.
- (V) So $3^{k+1} \geq (k+1)^3$, and the result follows for every n by repeating the step.

Which of the following is correct?

- A The argument is valid and the claim is true.
- B The argument is invalid, and the claim is false for some n .
- C The claim is true, but line IV fails for $k = 1$ and $k = 2$, so the argument does not prove it as written.
- D The claim is true, but line III is invalid.
- E Line IV is false for every $k \geq 3$.
- F The argument would be valid if line I checked $n = 2$ instead.

Question 11

A function f is differentiable for all real x . Its derivative satisfies $f'(x) > 0$ for $x < 1$, $f'(1) = 0$, $f'(x) < 0$ for $1 < x < 3$, $f'(3) = 0$ and $f'(x) > 0$ for $x > 3$. It is also known that $f(1) = 2$ and $f(3) = -2$. Nothing else is known about f . Which of the following must be true?

- I The equation $f(x) = 0$ has a solution between 1 and 3.
 - II The equation $f(x) = 0$ has exactly three real solutions.
 - III The equation $f(x) = 2$ has exactly two real solutions.
 - IV $f(0) < 2$
- A I and IV only
 - B I, II and IV only
 - C I, II, III and IV
 - D I only
 - E I, III and IV only
 - F II and III only

Question 12

The function f is continuous and increasing on $[0, 1]$, with $f(0) = 0$, $f(1) = 1$ and $\int_0^1 f(x) dx = \frac{1}{3}$.

Let f^{-1} be its inverse function. Which of the following must be true?

- I $\int_0^1 f^{-1}(x) dx = \frac{2}{3}$
- II $f(c) < c$ for some c in $(0, 1)$.
- III $f(\frac{1}{2}) \leq \frac{1}{2}$
- IV $f(x) \leq x$ for every x in $[0, 1]$.

- A I only
- B I, II and III only
- C I, II, III and IV
- D II and IV only
- E I and II only
- F I and IV only

Question 13

Let $f(x) = ax^2 + bx + c$, where a , b and c are real. Which of the following conditions are *sufficient* for $f(x) > 0$ for every x in the interval $[0, 1]$?

- I $a > 0$ and $b^2 < 4ac$
- II $f(0) > 0$ and $f(1) > 0$
- III $f(0) > 0$, $f(\frac{1}{2}) > 0$ and $f(1) > 0$
- IV $c > 0$, $a + b + c > 0$ and $b \geq 0$
- A I only
- B I, III and IV only
- C IV only
- D I and IV only
- E I, II, III and IV
- F III and IV only

Question 14

Consider cubic equations $x^3 + px + q = 0$ with real p and q . Which of the following conditions are *sufficient* for the equation to have three distinct real roots?

- I $p < 0$
- II $p < -3$ and $|q| < 2$
- III $q = 0$ and $p < 0$
- IV $p < 0$ and $q > 0$
- A II only
- B II and III only
- C I, II and III only
- D III only
- E II, III and IV only
- F I, II, III and IV

Question 15

A student claims that $x + \frac{1}{x^2} \geq 2$ for every $x > 0$, and argues as follows: “By the AM-GM inequality,

$x + \frac{1}{x^2} \geq 2\sqrt{x \cdot \frac{1}{x^2}} = \frac{2}{\sqrt{x}}$, and $\frac{2}{\sqrt{x}} \geq 2$.” Which of the following is correct?

- A The argument is valid and the claim is true.
- B The claim is true, but AM-GM does not apply to these two terms.
- C The claim is true, but the argument only proves it for $x \leq 1$.
- D The claim is true for all $x > 0$ except $x = 1$.
- E The claim is false: the argument fails because $\frac{2}{\sqrt{x}} \geq 2$ only for $x \leq 1$, and the least value of $x + \frac{1}{x^2}$ is less than 2.
- F The claim is false only for $x > 4$.

Question 16

For which of the following does there exist a function satisfying all the conditions stated?

- I A polynomial p with $p(0) = p(1) = 0$ and $p'(x) > 0$ for every x in $(0, 1)$.
 - II A polynomial p with $p(0) = 0$, $p(1) = 1$ and $p'(x) < 0$ for every x in $(0, 1)$.
 - III A continuous function f on $[0, 1]$ with $f(0) = 0$ and $f(1) = 1$ that takes the value 2 somewhere in $[0, 1]$.
 - IV A differentiable function f with $f(0) = 0$, $f(1) = 2$ and $|f'(x)| \leq 1$ for every real x .
- A none of them
 - B I and III only
 - C III only
 - D III and IV only
 - E I, III and IV only
 - F I, II, III and IV

Question 17

Call a positive integer *representable* if it can be written as $5a + 7b$ with a and b non-negative integers. Which of the following statements are true?

- I The largest positive integer that is not representable is 23.
 - II Exactly 12 positive integers are not representable.
 - III Every multiple of 3 greater than 6 is representable.
- A I and II only
 - B I only
 - C I, II and III
 - D II only
 - E I and III only
 - F II and III only

Question 18

A function f from the real numbers to the real numbers satisfies $f(f(x)) = x$ for every real x . Which of the following must be true?

- I f is one-to-one.
 - II f is increasing.
 - III If f is increasing, then $f(x) = x$ for every x .
 - IV There is a real number a with $f(a) = a$.
- A I only
 - B III only
 - C I, III and IV only
 - D I, II, III and IV
 - E I and III only
 - F II and III only

Question 19

For which of the following combinations does there exist a function f , defined and twice differentiable for all real x , satisfying all three conditions for every real x ?

- I $f(x) > 0$, $f'(x) > 0$, $f''(x) > 0$
 - II $f(x) > 0$, $f'(x) < 0$, $f''(x) > 0$
 - III $f(x) > 0$, $f'(x) < 0$, $f''(x) < 0$
 - IV $f(x) > 0$, $f'(x) > 0$, $f''(x) < 0$
- A I only
 - B I, II, III and IV
 - C I, II and IV only
 - D I and II only
 - E II and III only
 - F I and IV only

Question 20

Let $f(x) = x^2 - 2$, and write $f^{(2)} = f \circ f$ and $f^{(3)} = f \circ f \circ f$. Which of the following are true?

- I $f^{(2)}(x) = x$ has exactly 4 real solutions.
 - II Every solution of $f(x) = x$ is a solution of $f^{(2)}(x) = x$.
 - III $f^{(3)}(x) = x$ has exactly 8 real solutions.
 - IV $f^{(2)}(x) = x$ has a solution with $|x| > 2$.
- A I and II only
 - B I, II and III only
 - C I, II, III and IV
 - D II and III only
 - E I only
 - F I and III only

END OF PAPER